

18.5.2016 (Fri)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
May - June 2016

Max. Marks: 100

Class: Second Year

Name of the Course: Database Management System

Course Code: UITC302

Duration: 3hrs

Semester: III

Branch: I.T.

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	List & explain responsibilities of Database Administrator.	10
Q 1 (b)	Draw & explain Database System Structure.	10
Q2 (a)	Draw ER diagram for University database consisting of four entities: Student, Department, Class and Faculty. Student has unique id. The student can enroll for multiple classes. Faculty belongs to department and faculty can teach multiple classes. Each class is taught by only one faculty. Every student will get a grade for class he/she has enrolled.	10
Q2 (b)	Write EER features with example.	10
Q3 (a)	Consider following schema and write SQL queries for the following: Employee(<u>ename</u> , city, date_of_joining) Works(<u>ename</u> , cname, salary) Company(<u>cname</u> , city) 1. Create employee table with primary key constraint. 2. Modify the database so that John now lives in 'Mumbai'. 3. List employees who live in the same city as their company.	10
Q3 (b)	Convert the relation R(P,Q,R,S,T,U) with following functional dependencies into 2NF and 3NF. $P \rightarrow Q$ $ST \rightarrow PR$ $S \rightarrow U$	10
Q4 (a)	Explain Trigger with an example. OR Explain Security and Authorizations with an example.	10
Q4 (b)	Distinguish between Static hashing and Dynamic Hashing. OR Explain steps for Query Optimization.	10
Q5 (a)	Explain Serializability with an example. OR Explain Timestamp based protocols.	10
Q5 (b)	Write short note on: (any two) 1. Shadow Paging 2. ACID Properties 3. Log Based Recovery 4. Deadlock Handling	10

